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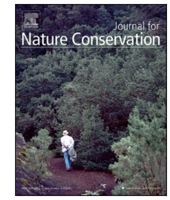

Habitat management carried out by hunters in the European turtle dove western flyway: Opportunities and pitfalls for linking with sustainable hunting

Carlos Sánchez-García [a,*], Thibaut Powolny [b,c], Hervé Lormée [b], Susana Dias [d], Francesc Sardà-Palomera [e], Gerard Bota [e], Beatriz Arroyo [f]

[a] *Fundación Artemisan, Department of Research, Ciudad Real 13001, Spain*
[b] *Office Français de la Biodiversité, Direction de la Recherche et de l'Appui Scientifique, 79360 Villiers-en-Bois, France*
[c] *Fédération Départementale des Chasseurs du Doubs, route du Chatelard, 25360 Gonsans, France*
[d] *Centro de Ecologia Aplicada Baeta Neves, Instituto Superior de Agronomia, Universidade de Lisboa (CEABN/ISA, UL); INBIO, Research Network in Biodiversity and Evolutionary Biology – InBIO, Tapada da Ajuda, 1349 Lisboa, Portugal*
[e] *Conservation Biology Group, Landscape Dynamics and Biodiversity Program, Forest Science and Technology Centre of Catalonia (CTFC), Solsona, Spain*
[f] *Instituto de Investigación en Recursos Cinegéticos (IREC, CSIC-UCLM-JCCM), Ronda de Toledo 12, Ciudad Real, Spain*

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ABSTRACT

As described in many farmland birds of Europe, habitat loss is one of the main factors explaining the decline of the European turtle dove (*Streptopelia turtur*), a migratory columbid which breeds in the Western Palaearctic and winters in Africa. In the countries belonging to the western flyway (which host around 40–50% of the European breeding population), the species was hunted before the implementation of a moratorium in recent years, and it is known that habitat management is conducted to favour game and wildlife. Through a questionnaire we investigated the habitat management conducted in hunting grounds of France, Portugal and Spain (931 respondents), and the measures that may favour turtle doves. The vast majority of respondents were totally or partially in charge of game management (98%), but less than 50% were responsible for decisions regarding farming or forestry. The most frequent measures were supplementary food (73% of respondents), water provision (68%), forest management (50%) and agricultural management (47%). Land interventions were conducted in a high proportion of grounds, but at limited scale (<50 ha/hunting ground). In most cases, the management was self-funded by hunters and game managers. Habitat management measures were rarely targeted exclusively to the turtle dove; interventions were, in contrast, frequently part of management targeting various species simultaneously. Grounds with a tradition of higher number of hunted doves were those more likely to target the species in their management. The likelihood of implementing management tools was mainly linked to variations in hunting pressure rather than to perceptions of turtle dove population trends. As the turtle dove is favoured by the ecotone between woodland and farmland, it is crucial to promote and reward measures beyond food and water provision, involving agricultural and forest management, ideally accompanied by species' monitoring in hunting grounds. As the current Adaptive Harvest Management Framework promoted by the European Union aims to link hunting opportunities with habitat enhancements, hunting grounds could be crucial to improve turtle dove habitats at large scale and help reversing current population trends.

1. Introduction

The European turtle dove (*Streptopelia turtur*) is a widely distributed breeding bird in Europe, North Africa and Asia (Cramp, 1985). In Europe, it migrates to the wintering quarters located in sub-Saharan Africa through two main migratory flyways, one passing through the Iberian Peninsula and western Mediterranean sea (hereafter the western flyway) and another one passing through Italy and Greece (hereafter the central-eastern flyway) (Marx et al., 2020). The European breeding population is estimated at 3.1–5.9 million pairs, of which 40% are

* Corresponding author at: Fundación Artemisan, 13001 Ciudad Real, Spain.
E-mail address: investigacion@fundacionartemisan.com (C. Sánchez-García).

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located in Spain (BirdLife International, 2019). During the last decades, the European turtle dove (hereafter TD), has experienced a sharp population decline, particularly in the western flyway, which has been mainly attributed to habitat loss and deterioration on both the breeding and wintering grounds, illegal killing and unsustainable hunting levels (Fisher et al., 2018). The species is now classified as vulnerable on the European Red List of Birds (BirdLife International, 2019). In 2018, an International Species Action Plan (ISAP) was approved in the European Union (EU), aiming to halt and reverse the TD population decline (Fisher et al., 2018). Among its objectives, the ISAP aims to maintain or increase good quality habitats on the breeding grounds and ensure that hunting across the range of the EU member states is conducted at sustainable levels. In this context, in 2019 an international Adaptive Harvest Mechanism was initiated in the EU (tender ENV.D.3/SER/2019/0021), which overall aimed to base hunting decisions on outcomes of a population model, and also to promote and reward efforts by hunters in implementing habitat management measures. In 2021 and 2022, a temporary hunting moratorium was established in the western flyway as part of that AHM, but when the conditions to resume hunting are met, there is an objective to link hunting opportunities to the implementation of habitat management measures by hunters (Arroyo, 2022). It has frequently been mentioned that maintaining hunting activities of the TD to some extent could be a good incentive to maintain and increase the efforts by hunters to improve habitat quality for the species in Europe (Arroyo, 2022; Lormée et al., 2020). A better understanding of what management is conducted in hunting grounds, which of the implemented tools can favour TD, and what are the barriers for further implementation of habitat management may help to develop tailored measures for the species, both at farmland and forest habitats.

The species favours the ecotone between farmland and woodland, i.e. mixed habitats which provide food and nesting substrates (Carboneras et al., 2022), and in France it has been demonstrated that abundance increases with fallow lands and hedge availability (Sauser et al., 2022). It is also a strict granivorous species that starts breeding just after migration, and favours small grain of wild plants within its diet (Gutiérrez-Galán & Alonso, 2016; Carboneras et al., 2022; Young et al., 2023). Therefore, actions like favouring growth or maintenance of wild grasses (either in patches or strips along hedges), particularly early-flowering wild plants with a high production of seeds, maintaining woodland patches within farmland, or maintaining woodland clearings to avoid encroachment within woodland, are likely to be positive for the species (Carboneras et al., 2022). Some of these actions are partially similar to the ones implemented for other small and big game species. Consequently, management for small game species may include measures favouring TDs during the breeding season in Europe, even if that management does not primarily target TD.

For example, management targeting red-legged partridges in Iberian habitats can provide feeding habitats (field margins, crops for wildlife, etc.), or nesting habitat (shrubs and forest management) for TDs (Sánchez-García et al., 2017). Supplementary grain and water are provided for partridges (both wild and farm-reared) in a significant proportion of hunting grounds (Arroyo et al., 2012), which may also benefit TD. "Management packages" targeting rabbits and hares (Ferreira et al., 2014; Lázaro et al., 2023), or implemented for wild pheasants (*Phasianus colchicus*) and grey partridges (*Perdix perdix*) in Europe, often include the implementation of cover crops, field margins and hedgerow management (Aebischer & Ewald, 2010; Draycott et al., 2002; Faragó et al., 2012), which may be beneficial for TD. Similarly, management targeting big game in Spain such as the red deer (*Cervus elaphus*) and roe deer (*Capreolus capreolus*) includes forest and crop management (Sáenz de Buruaga & Carranza, 2008) which favours wild seed availability for the TD (Gutiérrez-Galán & Alonso, 2016; Gutiérrez Galán, 2016).

Hence, a large range and diversity of habitat management measures exists, which might potentially impact in a positive way on the conservation and recovery of TD (Fisher et al., 2018). However, there are few synthetical studies that list the different types of habitat management performed, the spatial scale over which they are implemented, and whether these measures target specifically the TD or other species. More specifically, the few studies addressing the management of TD breeding habitat in Europe all refer to actions conducted only in south-western Spain, and describe mainly food provisioning actions (Rocha et al., 2009; Rocha & Hidalgo, 2002).

Accordingly, in this study, we used information from the 3 main TD hunting countries in the western flyway, Spain, Portugal and France, with the aim to (1) describe and quantify the management tools conducted in hunting grounds that may potentially benefit TD, at the scale of the western flyway, (2) identify whether the TD is perceived as a target species of management tools that may potentially benefit the species, and (3) understand which factors explain the incentive for hunters to conduct any type of management (including the perceptions of TD population trends or the hunting interest in the species), and alternatively what are the potential obstacles for implementing such efforts.

2. Materials and methods

2.1. Study area and context

This study was conducted in 3 countries (France, Spain, Portugal) included in the western flyway (Lormée et al., 2020; Lormée & Carboneras, 2021), that host around 40–50% of TD breeding population in Europe (Fisher et al., 2018). In all these countries hunters consider small game hunting as an important recreational activity and in some regions as a significant economic resource (for example the red-legged partridge in Spain and Portugal, see Pavia et al., 2017), or the common pheasant in France (Chardonnet et al., 2002). TD hunting has occurred historically in these three countries. For example, official data available for the 2013–2014 hunting season indicates that more than 1 million TD were harvested that year in those three countries, with Spain being the country with the highest share, followed by Portugal and France (Lormée et al., 2020). TD hunting mainly occurs from mid-August to late September (Fisher et al., 2018), depending on the country.

2.2. Questionnaire on habitat management in hunting grounds

A questionnaire was developed in order to collect detailed information about the habitat management conducted in hunting grounds that might benefit TD. The questionnaire was designed based on existing information about habitat management targeting game species, and face-to-face and phone interviews were conducted in Spain to hunters' federations, hunting associations, game managers and landowners to gain practical knowledge, as well as literature on small and big game species habitat management in Mediterranean contexts (Sáenz de Buruaga and Carranza, 2008). Given that several of the implemented management tools used might potentially benefit non-target species, including TD according to scientific information (Armenteros et al., 2021), and that not all participants would necessarily consider TD as a primary "target species", we designed a questionnaire addressing game management in general, in which managers could specify which species were targeted. The questionnaire had a combination of open and close questions (Supplementary material 1 and 2). In Spain and Portugal, before delivering, the questionnaire was piloted with 17 game managers to ensure correct wording. Pilot tests showed that the questionnaire took 15–20 min to be filled completely.

The questionnaire was implemented in Google Forms and distributed online through hunters organizations, regional governments and other organizations related to hunting, who made contacts available. Filling the questionnaire was voluntary, there was no predefined selection of respondents and it was not always possible to accurately assess the proportion of contacts who refused to fill the questionnaire. Owing to administrative differences among countries, the recipients of the questionnaires were slightly different. For example, in France, questionnaires were sent to Departmental Hunting Federations (*Féderations Départementales des Chasseurs*) and responses thus refer to all hunting estates within the administrative department. In Spain and Portugal, questionnaires were filled directly on-line by managers of individual hunting grounds, which received them through the regional hunting federations and other organizations, and also through some regional governments (Supplementary material 3). In all cases, the questionnaire included an initial statement explaining the objectives of the study and informed that all replies would be treated anonymously (i.e. only at regional level or per category of hunting state). All participants had to indicate explicitly they agreed to participate in the survey, and the final version of the questionnaire was reviewed and approved by the organizations and regional governments who distributed it. The questionnaire included questions about the following aspects:

1) Basic data related to the hunting ground (for Spain and Portugal) or the huntable area in the Department (France): location and surface, responsibility of the respondent about hunting, agricultural or forest management, type of hunting practice applied in the area of concern (including whether they hunted TDs and average annual take).
2) Nature of habitat management actions implemented: supplementary feeding, water provision, creation or maintenance of ponds, agricultural management for game and wildlife (including game crops, management and conservation of stubbles or fallows), existence of agro-environmental measures funded by public schemes (AEM, including measures similar to those described above), forest and shrub management (including forest or shrub clearing). In all these cases, respondents were asked whether each type of management was implemented or not, the total surface involved, who funded the management, and which were the game species targeted by the management.
3) Perceptions about changes over time about habitat and turtle dove populations in the hunting ground/area.
4) Perceptions about future possibilities for carrying out additional habitat management, and the type of obstacles for not implementing additional management.

The study complied with data protection law. The survey was conducted from June 2020 to June 2021. Most questionnaires were completed and returned in the course of 2020; however, owing to the low number of responses from Portugal that year, the study period was extended until 2021 in that country only.

2.3. Data analysis

Differences in proportions were tested with Chi-square tests. General Linear Models (GLM) were used to analyse whether the likelihood of specifying TD as a target species of habitat management depended on hunting pressure (set as a continuous variable, based on the annual take in the hunting ground averaged over the 2014–2019 period) and the perceptions of TD trends (a categorical variable with three levels: increase, decrease or stability). The response variable was fitted to a binomial error with a logit link function. Similarly, we also used GLMs to analyse factors influencing the likelihood of implementing different management measures in hunting grounds. Response variables (occurrence or not of a certain management tool in a hunting area) were also fitted to a Binomial error (with a logit link function). Explanatory variables included a continuous variable (hunting pressure in the period 2014–2019, as the average annual hunting bag or take) and five categorical variables: country (three levels: Portugal, France and Spain), hunting pressure change (three levels: increase, stable, decrease; this was calculated based on the comparison between data reported in the questionnaire about the hunting pressure 15 years ago and in the period 2014–2019), the perception of TD population trends in the last 5 years (three levels: decrease, increase, stable), whether the respondent was responsible of agricultural management, and whether they were responsible of forest management (three levels each: totally, partially or no). Type III results of the explanatory variables are presented. Analyses were carried out with the program R 4.1.1.

3. Results

A total of 931 valid questionnaires were returned: 679 from Spain, 197 from Portugal and 55 from France. As mentioned above, in Spain and Portugal, questionnaires were filled at the hunting ground level, while in France they were filled-in at the administrative Department level. Hence, responses covered a large part of the territory in France (56.5% of metropolitan France, Supplementary material 4). In Spain, responses came mainly from the regions of Andalucia, Castilla la Mancha, Castilla y León, Extremadura and Catalonia, regions that cover around 66% of the total Spanish land. The response rate varied among countries: in France, 57% of Departments provided a response. In Spain, overall response rate was 7% (i.e., 7% of the hunters contacted by the Regional Hunting Federations and Regional Governments that distributed the questionnaire), and pooling hunting grounds, they represented a minimum of 1.24 million hectares (2.3% of the total surface declared as "hunting ground"). In Portugal, responses were more concentrated in the Alentejo region (40% of the total), and the response rate was 3.9% (i.e. 3.9% of contacted managers provided a reply covering 228 hunting grounds), corresponding to 7% of the total area under hunting regulation).

3.1. Characteristics of respondents: Management responsibilities and TD hunting

In France, respondents were professional technicians employed by the hunting federations. In Spain and Portugal, respondents were usually the managers of the hunting grounds, which were not necessarily professional technicians, but hunters temporarily in charge of the running of the hunting grounds. Virtually all respondents from the three countries were totally or partially in charge of game management decisions (98%). In contrast, only 45% and 41% were responsible for decisions regarding farming or forestry respectively in Spain and Portugal, whereas in France, all respondents were at least partially responsible of both activities. Hunting grounds of 52% of respondents employed full or part-time gamekeepers, the proportion being higher in Portugal (56%) and lower in France (40%).

TD hunting occurred in the majority of hunting grounds of respondents. This was the case for 70% of respondents in Spain ($n = 668$), 86% of respondents in Portugal ($n = 197$) and 85% of respondents in France ($n = 55$), specified that TD was hunted in their area of concern.

3.2. Types of implemented management measures and source of funds for their implementation

The most frequent management measures implemented were provision of supplementary food (73% of respondents, $n = 931$). Supplementary food was either provided through feeders (31% of hunting grounds/departments in France), spread in plots (24%) or a combination of both (39%). Taken all together, 84% of hunting grounds provided food in summer, and around 35% all year long. In Spain and Portugal 98% of hunting estates provided water in summer and 40% all the year long. Creation or maintenance of ponds for wildlife (69%, $n = 930$), and provision of supplementary water in water troughs (67%, $n = 931$) were also widely used measures.

Forest management (shrub or tree clearings) was also carried out by 50% of respondents ($n = 928$), whereas 47% of respondents ($n = 813$) carried out some kind of agricultural management in their area. The most common agricultural management mentioned was the implementation of game crops ($n = 354$) or management of fallows ($n = 280$). AEM were mentioned to be implemented in 140 of the responses. The frequencies for each tool varied among countries (Fig. 1, $X^2_{10} = 83.6$, P <

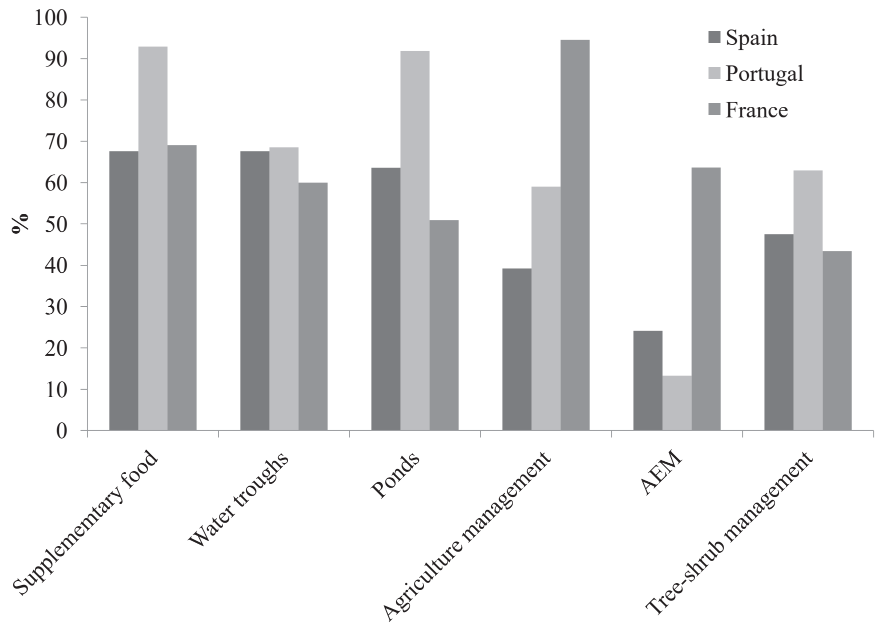

Fig. 1. Percentage of hunting grounds conducting the main management measures per country (AEM = agri-enviromental measures).

0.001): supplementary food, ponds for wildlife and forest management were more common in Portugal, whereas agricultural management was more common in France. It is worth noting that in the latter country respondents were more commonly, at least partially, in charge of agricultural management, and each response applied to a larger number of hunting grounds, so the likelihood of responding positively was higher, in any case.

About agriculture management, 33.4% of hunting grounds in Spain (n = 234) had only game crops whereas 59% declared combinations of game crops, management of fallow land and postharvest stubble maintenance; combinations of those measures were even more common among departments carrying out agriculture management in France (63.5%, n = 52) or among estates implementing agriculture management in Portugal (93.7%, n = 95) (Fig. 2). Looking at the surface of crops area that were managed, the majority of hunting grounds in Spain and Portugal (73–88%) managed crops for wildlife in 1–50 ha, whereas in France the majority of responses (73%) declared managed crops covering more than 50 ha.

40% of respondents (n = 930) also mentioned to implement other type of management measures. In France, these measures mainly included hedge planting (mentioned by 44 out of 55 respondents), whereas in Spain and Portugal they included a variety of measures, including planting shrubs or trees (n = 78), cleaning and maintenance of natural fountains or streams (n = 48), or tools not related to habitat management such as predator control (n = 13) or providing antiparasitic treatments for game (n = 10).

In most cases, hunters and game managers self-funded the management conducted. This was predominantly the case for supplementary

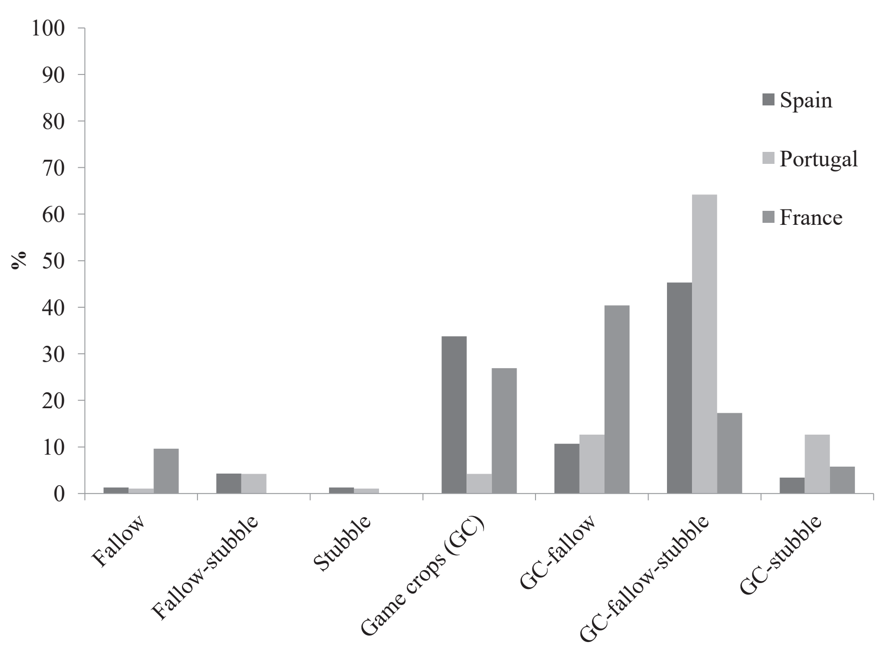

Fig. 2. Percentage of respondents conducting agricultural management (and combinations of different measures) per country.

food and water (98%, n = 680 and 523 respectively), and also when considering crop management for wildlife (game crops: funded by hunting grounds in 86% of cases, n = 352; stubble management, in 88% of cases, n = 198; fallow management, 85% of cases, n = 249). Forest/shrub clearing for wildlife was also funded by hunters (or hunters' associations) in a significant proportion of grounds of Spain (70%, n = 308), less in Portugal (41%, n = 120), where this tool was more often supported by public subsidies or funded directly by the forest owners. In France, hedge plantation was also funded by hunters. The only tool fully supported by public agencies was AEM. In France, agriculture management was conducted at a larger spatial scale than Spain and Portugal. In the hunting grounds, around 48% of the agricultural management measures were funded by hunters, 42% both by hunters and departmental/regional subsidies, and the remaining 10% only by public subsidies.

3.3. Turtle dove as a target species of implemented management

TD was mentioned as one of the target species for at least one of the management tools implemented by 87% of the respondents in Portugal and by 59% of the respondents in Spain. In France, most respondents answered that "all game species" at large were the target of management, so it was not possible to quantify for this species separately. On the other hand, only 29 respondents in Spain and Portugal mentioned the TD as the only target species of the management tools implemented. Hence, most of the time management tools aimed to benefit TD together with other small and big game species. In general, respondents mentioned more frequently TD as one of the targets of the management tools when the management focused on small game species in general than when it targeted also big game species (Fig. 3).

3.4. Perceptions about TD trends and future management possibilities

Perceptions about TD population trends varied depending on the period considered. In general, the majority (78%, n = 876) considered that there were fewer TD now than 20 years ago, but the proportion was lower when considering changes over the last 5 years, where only 60% (n = 931) perceived a decline and 33% considered that the population had stabilized in that period.

Perceptions in Spain and France depended also on hunting intensity (defined as the size of the annual hunting bag or take) in the hunting estate: a population decline in the last 20 years was more frequently perceived in hunting grounds that had higher hunting bags 15 years ago, whereas those with low hunting bags during the same period more frequently perceived that TD populations had been stable in the last 20 years ($X^2_8 = 40.92$, $P < 0.001$, Fig. 4a). On the other hand, hunting grounds showing higher hunting bags in a more recent period (2014–2019) perceived more frequently that populations were stable or increasing, whereas hunting grounds with lower hunting bags in recent years more frequently perceived that TD populations had recently decreased ($X^2_8 = 40.39$, $P < 0.001$, Fig. 4b). In general, respondents that perceived that the population had decreased were more likely to have reduced hunting bags in the last 15 years, whereas those perceiving that populations had increased were more likely to have increased hunting bags during that time ($X^2_4 = 146.23$, $P < 0.001$, Fig. 4c).

When asked about whether more actions could be made to favour TD, 58% (n = 931) answered positively. Management measures mentioned primarily in this context included providing more game crops (29%, n = 275), together with food or water provision (17%, n = 163), and management of shrubs, hedges or trees (6%, n = 56). Mentioned obstacles for implementing further management were limitations due to not owning land (48%, n = 449) and economic limitations (38%, n = 355).

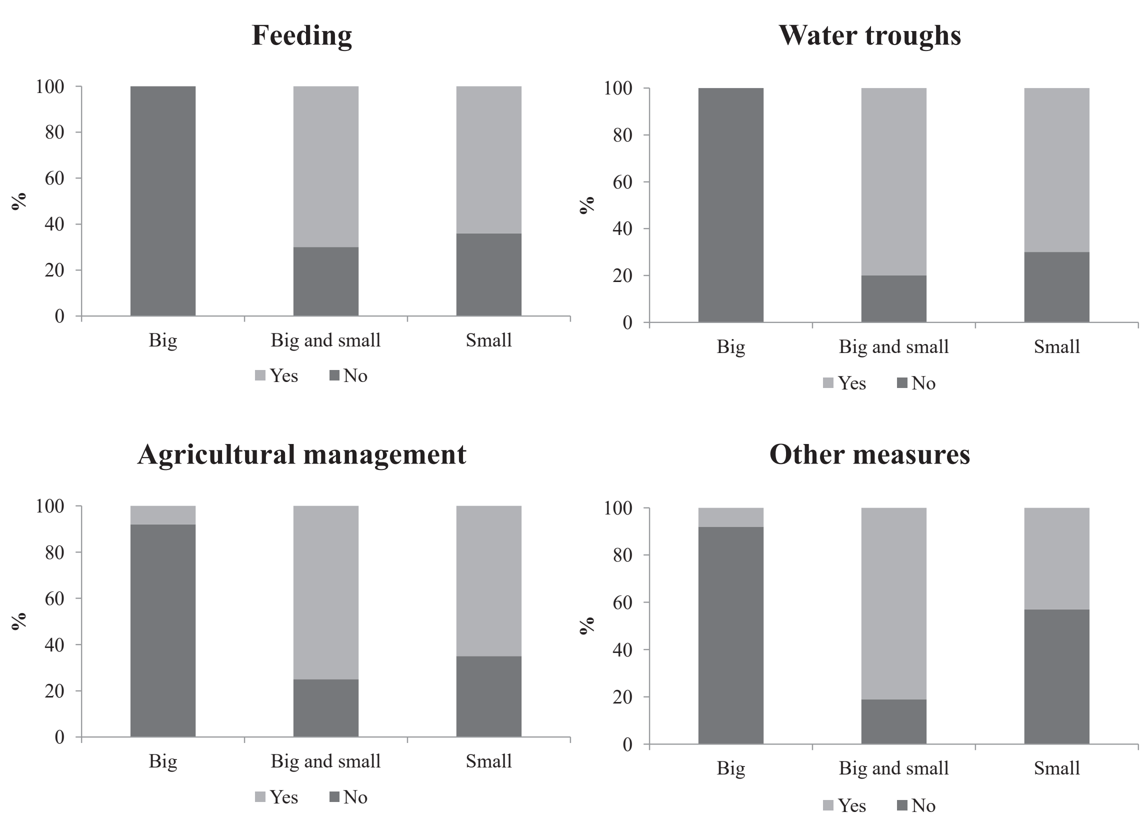

Fig. 3. Percentage of respondents declaring to target TD with their management, considering the type of measure and other game species targeted. These data include only responses from Spain and Portugal, as most French respondents did not include TD specifically in their responses.

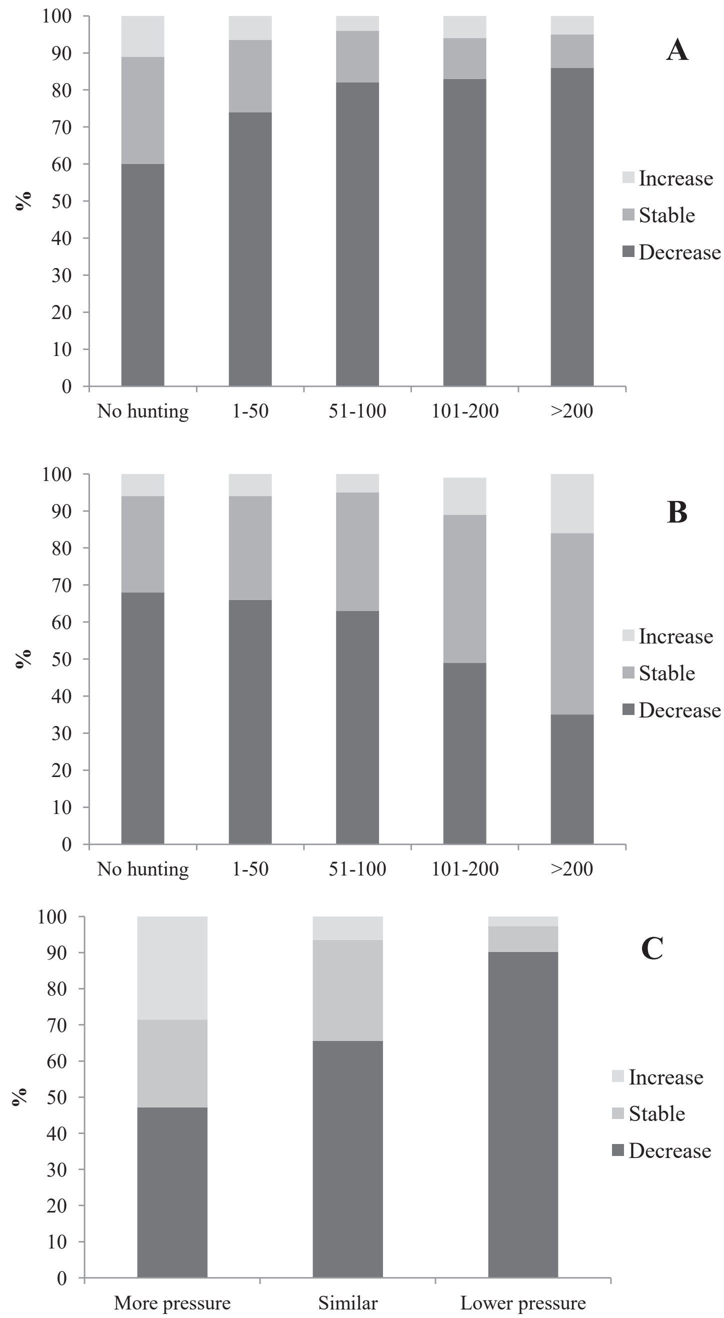

Fig. 4. Relationships between perceptions of TD population trends and hunting intensity (average number of turtle doves shot per year in the hunting ground). A) Trends in the last 20 years and hunting intensity 15 years ago; B) trends in the last 5 years and hunting intensity 2014–2019; C) TD population trends and change in hunting intensity. These data include only responses from Spain and Portugal, as most French respondents did not include data about TD harvest.

3.5. Relationships between TD-targeted management, TD hunting interests and population trends perceptions

Considering only Spain and Portugal (the only two countries where TD was sometimes specifically mentioned as a target of management), the likelihood of TD being mentioned as a specific target of management increased in hunting grounds with higher TD hunting bags ($X_4^2 = 74.96$, $P < 0.001$, Fig. 5a). It is worth mentioning that even when no hunting was conducted, almost half of the grounds considered the TD as a target species (Fig. 5a). Additionally, the perception about the population trends of TD in recent decades affected whether the species was targeted or not by the implemented management: the TD was more likely to be specifically mentioned as a target species in hunting grounds with positive perceptions of TD population trends ($X_2^2 = 11.34$, $P = 0.003$, Fig. 5b). In a GLM assessing the likelihood of TD being specifically targeted in management, both variables were additively significant (TD trends, $Chi_2^2 = 6.14$, $P = 0.046$, hunting pressure: $Chi_1^2 = 59.4$, $P < 0.001$).

3.6. Factors influencing the likelihood of using different management measures

We found various factors significantly influencing the likelihood of implementing management actions (Table 1). The probability of providing supplementary food varied among countries, and within Spain and Portugal, with hunting pressure and hunting pressure change: it was significantly higher at grounds where TD was hunted in higher numbers in recent years, and where hunting pressure had increased over

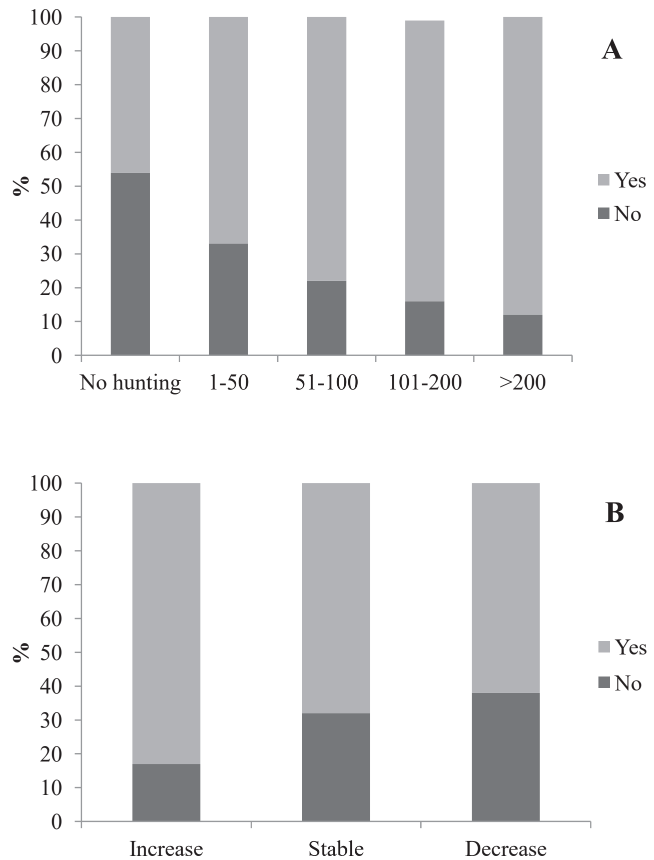

Fig. 5. Percentage of respondents mentioning to target TD with their management, in relation to A) their TD harvesting levels for the period 2013–2019 and B) their perception of TD population trends. These data include only responses from Spain and Portugal, as most French respondents did not include TD specifically in their responses and did not include data about TD harvest.

time. Additionally, it varied with the perception of TD trends, being higher in hunting grounds where there was a perceived population increase (Table 1, Fig. 6). The likelihood of providing supplementary water also varied with hunting pressure and TD trend perception (Table 1, Fig. 6). In contrast, the likelihood of implementing crop or forest management only depended significantly on country and whether the respondent had the responsibility of agricultural or forest management in the hunting ground, respectively (Table 1, Fig. 6).

4. Discussion

Our study shows that a significant proportion of hunting grounds within the TD western flyway conducted management measures which may benefit the species, although those measures were frequently not exclusively targeted to the TD. Management decisions were linked to the hunting grounds' governance and hunting interest, and also to perceptions on TD abundance. As linking hunting opportunities with habitat enhancements is one of the aims of the Adaptive Harvest Management Framework (Arroyo, 2022), our results could help promoting tailored management for the species and contribute to sustainable hunting once the moratorium is lifted, and also point to aspects that may be improved in order to maximize their effective implementation.

Despite differences on the unit of analysis among countries (hunting ground in Spain and Portugal, and *Département* in France), our results show the large scale in which management was conducted to improve habitat for game and wildlife. Supplementary feeding and water troughs were more frequent compared to agricultural management (mainly provision of game crops or maintenance or management of stubble of fallows for game), the implementation of specific AEMs and forest management, but the latter occurred also in a relatively large proportion of hunting grounds. Our results are in agreement with conclusions from grounds targeting small game species such as red-legged partridges (Ponce-Boutin et al., 2006; Arroyo et al., 2012; Reino et al., 2016), and wild rabbits (Ferreira et al., 2014), in which different combinations of management tools were implemented but dominated by provision of food, water troughs and secondarily by management of crops for game.

The higher frequency of supplementary food or water in relation to agricultural or forest management in the Iberian Peninsula could be explained by the fact that in Spain and Portugal less than 50% of hunting grounds had responsibility on farming and forest management. The higher proportion of respondents in France in relation to implementing AEM in France, could be related to the unit of analysis, the *Département*: as each includes many hunting grounds, it is more likely that one among them has at least some partial governance on agriculture and forest management. Indeed, all of the respondents indicated that at least partial responsibilities over those actions were held by hunters or hunting federations in their department. In other words, management is influenced by land ownership and governance: in the studied countries, the landowner holds the hunting rights, which may be leased to a person or a hunting association (Mustin et al., 2011). Thus, the land-management decisions belong to the landowner and hunters may struggle to take part in decisions regarding land use. In contrast, provision of supplementary food or water may be implemented by hunting organizations without holding rights on land use, as those may be put in common property areas (e.g. close to tracks, field edges, wastelands, etc). The fact that a significant proportion of the management implemented was funded totally or partially by hunters, highlights the large resources invested by the sector (Sánchez-García et al., 2021) which may benefit game and other wildlife (Caro et al., 2014), and the overall motivation of hunters to improve the conditions of game species. However, given that direct agricultural and farmland management are the management tools most likely to improve TD populations (in comparison to providing food or water, Carboneras et al., 2022), it would be particularly useful to develop appropriate systems to increase the possibility of hunters being directly involved in those activities, as this would increase the ecological impact of the hunters' investment.

The widespread use of food and water provision reported in the questionnaire agrees with existing literature indicating that in the Iberian Peninsula it is indeed a widespread tool targeting many game

Table 1
Type III results of the GLMs analysing the likelihood of implementing different anagement tools in hunting grounds (AEM = agri-environmental measures).

	Supplementary food			Water troughs			Crop management			AEM			Forest/shrub clearing		
	Chisq	Df	P	Chisq	Df	P	Chisq	Df	P	Chisq	Df	P	Chisq	Df	P
Country	50.17	2	**<0.0001**	0.42	2	0.81	34.77	2	**<0.0001**	14.02	2	**0.0009**	20.72	2	**<0.0001**
Responsibility of forest management	5.08	2	0.08	1.03	2	0.59	4.76	2	0.09	7.12	2	**0.028**	14.27	2	**0.0008**
Responsibility of agricultural management	2.48	2	0.28	4.8	2	0.09	9.69	2	**0.008**	1.71	2	0.42	2.32	2	0.31
Hunting pressure 2014–2019	12.77	1	**0.0003**	20.52	1	**<0.0001**	3.43	1	0.06	0.93	1	0.33	0.04	1	0.83
Hunting pressure change	19.19	2	**<0.0001**	1.27	2	0.52	1.05	2	0.59	0.2	2	0.9	0.02	2	0.99
TD trend perception	10.97	2	**0.004**	5.91	2	**0.05**	0.84	2	0.65	1.94	2	0.38	3.08	2	0.21

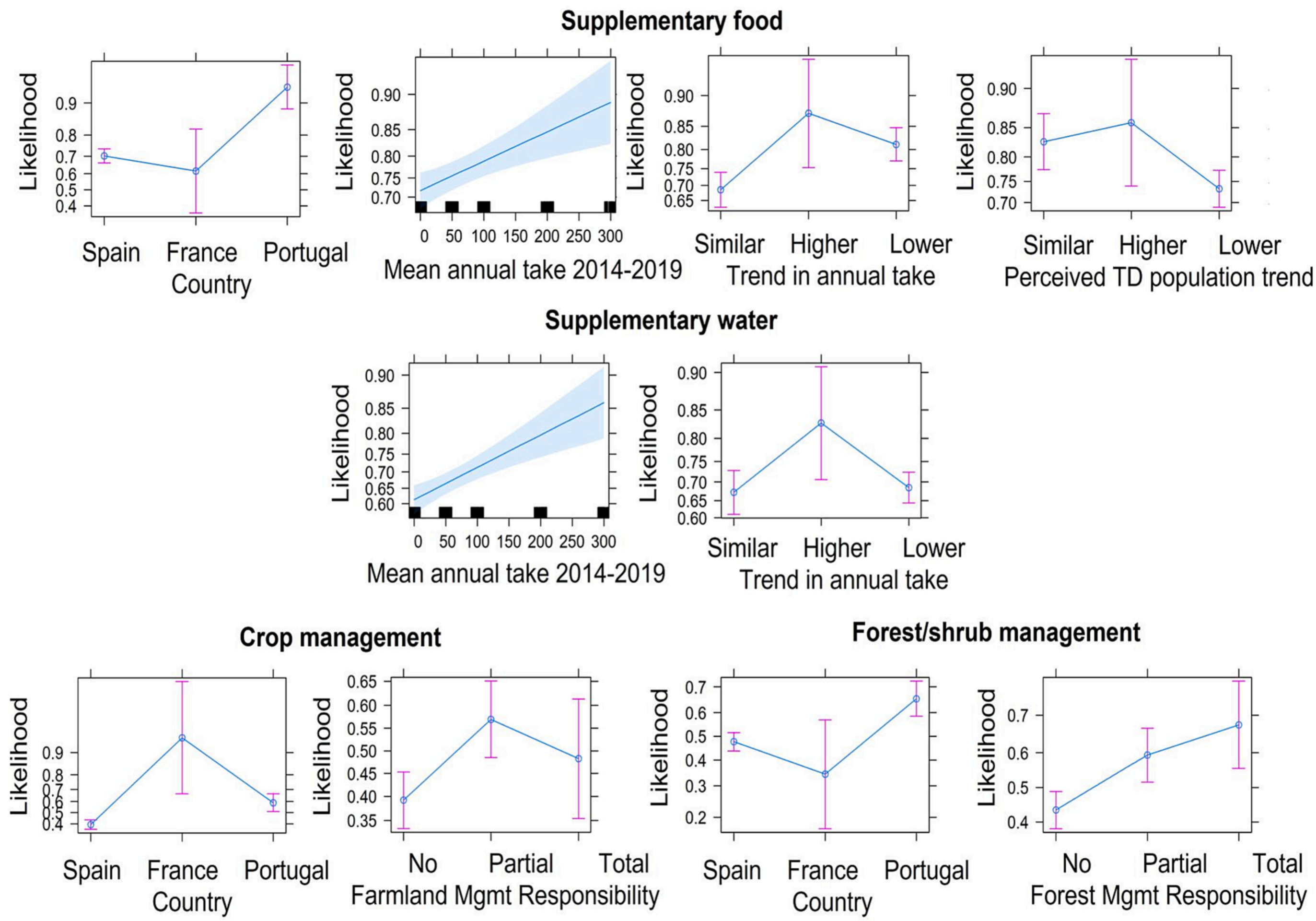

Fig. 6. Graphical outputs of the significant results of the GLMs analysing the likelihood of implementing different management tools.

species (Arroyo et al., 2012; Beja et al., 2009; Estrada et al., 2015; Sáenz de Buruaga & Carranza, 2008) including TD and other hunted columbids (Rocha et al., 2009), which may explain why the majority of grounds provided food during long periods of time (a third of them all year round). Provision of water is a measure of special importance in areas where drought periods are frequent (Lacasa et al., 2010). The ecological impact of supplementary food on TD populations is debated. Some studies suggest that the provision of supplementary food may help to increase productivity and recruitment of the TD, but that may also lead to overharvesting derived from bird aggregation at certain areas (Rocha & Quillfeldt, 2015; Rocha et al., 2022). A lack of relationship between supplementary food and improved breeding success has been found in yet other studies (Browne & Aebischer, 2003). Hanane et al. (2023), comparing managed and unmanaged hunting estates of Morocco, showed that food availability influences TD spatial patterns. Dunn et al. (2018) suggested that conservation strategies should include the provision of anthropogenic seeds for adults early in the breeding season, but they also showed that physical condition of nestlings was worse in those fed with crop seeds than in those fed with wild seeds (Dunn et al., 2017). Also, some authors question whether supplementary food indeed improves breeding success, and argue that rather than supplementary food provision, increasing wild seed availability would be a more efficient strategy (Carboneras et al., 2022), although results from ongoing studies cannot rule out that feeding may help to increase the carrying capacity of breeding TD populations (Arroyo, 2022). In the case of TD, and as specified in the Species Action Plan (Fisher et al., 2018), supplementary food should be considered as a “first aid” for TD populations where food scarcity has been shown to be a limiting factor, but should be implemented in combination with other measures to increase survival and breeding densities which would ultimately increase population abundance sufficiently to allow some sustainable hunting. More research is needed to address the effects of supplementary feeding of TD in different management contexts, also considering other factors such as habitat suitability and human disturbance.

Despite the lower governance capacity, it is noteworthy that agricultural management tools were conducted in a significant proportion of the grounds, especially in France (94% of respondents) and Portugal (64%), together with a lower (but non negligible) proportion of forest management. In the majority of grounds, a combination of different measures (game crops, management of fallow or stubble) was most common, although the surface of managed areas in Spain and Portugal was mainly below 50 ha in each hunting ground, indicating that the scope of these interventions was limited. Although we cannot rule out that respondents of the questionnaire were overall biased in favour of those more interested on TD (particularly in Portugal, where sample size was lower), the overall results for all countries confirm that land interventions are conducted in a high proportion of hunting grounds, but at limited geographical scale. Hence, the efforts conducted in the hunting grounds would not be enough to increase the ecotone between farmland and woodland at a landscape scale (Carboneras et al., 2022), which may hamper the recovery of the species. In that sense, the main barriers mentioned preventing more habitat management by hunters were funding and land use governance. If those two barriers could be overcome, habitat management for the recovery of TD could be largely increased through hunters or hunting organizations.

Our study also shows that the majority of grounds were involved in

TD habitat management, though the level of investment was influenced by hunting interest. Thus, hunting grounds showing higher number of hunted TD across time where those more likely to be targeting TD in their management. This supports the idea that maintaining the possibility of TD hunting could be a good incentive to maintain and improve the efforts by hunters to improve habitat quality for the species in Europe, but that this may be limited to provision of food and water (if the governance for land use is not modified within hunting grounds). Interestingly enough, TD remained a targeted species of management in almost half of the hunting grounds where no TD hunting was conducted, indicating that hunting is not an absolute precondition for management to improve the conditions for a species to occur. Thus, our results support (or at least do not contradict) that that low hunting levels determined by scientific monitoring could be associated with management at large scale, as long as investment conducted by hunters is also complemented by public funding. Our results also show that shooting restraint is part of management decisions (as implied by grounds implementing management for TD but not hunting), though it is unknown whether the current hunting moratorium (and its length) may affect the level of management targeting columbids in general. Interestingly, in the Iberian Peninsula only a small proportion of the grounds devoted to big game considered the TD as a target species of their game management, likely because TD hunting does not frequently occur in big game hunting estates, even when carrying out management tools likely to be beneficial for the species. Additionally, big game hunting grounds (which usually have a higher proportion of woodland) may offer better habitat conditions compared to small ones, owing to the optimal nesting habitat and higher wild seed provision compared to small game ones, key factors for breeding TD populations in Iberia (Carboneras et al., 2022; Gutierrez-Galán et al., 2018). It would be important to better convey to hunting managers the potential positive impacts of their management to non-target species (including TD), so they may optimize their implementation even in the absence of a direct hunting interest in the species.

On the other hand, analyses of the likelihood of each management tool separately showed that hunting interests mainly influence the rate of feeding and water provision, but not of agricultural of forest management. Thus, the probability of providing supplementary food was significantly higher at grounds where TD was hunted in higher numbers in recent years, and where hunting pressure had increased over time. The latter also points to supplementary food leading to bird aggregation at certain areas, which may help sustain high hunting bags (Rocha and Quillfeldt, 2015; Rocha et al., 2022), potentially leading to overharvest as local population at hunting time may be unrelated to breeding numbers. In that sense, the provision of supplementary food may alter the perception of population abundance. In that context, it is also relevant to highlight that the use of supplementary food was related to perceived population increases. This, on the one hand, could be an indication that local population indeed increases as a consequence of the provision of supplementary food, as suggested by some studies (Rocha et al., 2022). On the other hand, it could be an indication that providing food "mitigates" the perception of population declines, as turtle doves aggregate around the feeding plots in late summer (Rocha & Quillfeldt, 2015), leading to potentially inaccurate information about population abundance. Indeed, until recent times the most widespread method to monitor TD abundance in hunting grounds was to count birds in summer at food plots (Rocha et al., 2022), which precludes comparisons with the information obtained through traditional bird monitoring programs. Once enough information is gathered through monitoring TD abundance in spring, as recently implemented in many hunting grounds through voluntary schemes (such as www.observatoriocinegetico.org in Spain), it will be possible to discriminate between both hypotheses.

In any case, our results also point to the fact that perceptions of TD trends were related to hunting levels: long-term population declines were more frequently perceived in hunting grounds that used to have large bags 15 years ago, possibly because such large bags are currently very rarely experienced (Moreno-Zarate et al., 2021). In contrast, hunting grounds with higher hunting bags in recent years perceived more frequently that populations were stable or increasing, possibly because those are the ones using supplementary food and where TDs aggregate in summer, so local abundance just prior to the hunting season may have been maintained, even if breeding populations decline. Hunting grounds where the perception was that the TD population had decreased were more likely to have reduced hunting bags in the last 15 years, whereas those perceiving that populations had increased were more likely to have increased hunting bags during that time. This association either points towards the idea that TD hunting decisions are indeed linked to field perceptions, or that the perception of the population trends is mainly based on variations in hunting success (i.e., "*if it is more difficult to capture as many turtle doves as before, then there must be fewer, whereas if the hunting success is similar, this points to numbers being similar*"). For a migratory species that has important pre-migratory movements, local abundance at hunting time may or may not be a good indicator of overall breeding numbers, and hunting bags at the local level are related to a variety of variables that may be independent of breeding numbers (Delibes-Mateos et al., 2021). It would be particularly important to ascertain and quantify the relationship between breeding numbers, hunting pressure and hunting management to be able to better adjust perceptions, hunting decisions and management decisions. In that context, initiatives to involve hunters in direct monitoring of breeding numbers are likely to be particularly constructive with that aim.

Considering our results, what are the opportunities and pitfalls to link habitat management with hunting opportunities for the TD? First, it is clear that hunters conduct habitat management and invest large resources to improve habitats for game species, so they are key stakeholders for the recovery of TD populations. However, hunters' efforts are limited by both funding and the possibility of acting (not having responsibilities over land use in many hunting grounds). Therefore, hunters' efforts should be complemented with the Common Agriculture Policy (CAP) and go beyond areas where hunting takes place, because current measures may not be efficient to conserve or allow improvements of TD populations at large scale. As shown in results, the surface of agricultural and forest management in hunting grounds managed by hunters is currently rather limited, hence interventions to increase habitat complexity and heterogeneity beyond those already implemented should be promoted, as well as finding administrative methods and funds to further increase the efforts made by hunting organisations.

Secondly, many management tools conducted at the hunting grounds (regardless of whether TD is a specific target or not) are likely to be beneficial to the TD because they may increase food supply (e.g. shrub clearings, game crops, supplementary food) or nesting habitats (hedge plantings, tree plantings, etc). However, there is insufficient scientific evidence of whether specific habitat measures are providing key requirements for the species, or their relative relevance for the recovery of the species in different contexts. It may be that some measures should be promoted, while other diminished or "reshaped", especially when considering that TD requirements may vary depending on habitat types and regional and national contexts. Further research is needed to address whether current measures fit within the TD needs (Carboneras et al., 2022).

Lastly, and particularly relevant given all the things mentioned above, it is important to improve TD monitoring, both in hunting grounds (where monitoring protocols should follow scientific advice) and overall. In that sense, it would be useful if hunters could further contribute to the common bird monitoring protocols that are used to assess the flyway-level population trends, adding information from further areas. Additionally, hunting-ground monitoring (e.g. www.observatoriocinegetico.org) may help to evaluate the efficiency of different management tools and contribute to decisions on sustainable hunting (quotas).

5. Management implications for an Adaptive Harvest Management scheme linked to habitat management

As part of the current AHM for the TD, one of the tasks is to develop mechanisms to identify and reward habitat management conducted by hunters. Given our results, it would be particularly relevant that the scheme prioritizes management tools beyond the provision of supplementary food or water, given that those two management tools are widespread for all game species. Therefore, hunting grounds promoted and rewarded should be those that include, in addition to food and water provision, agricultural or forest management that is likely to be beneficial for the species. For the mechanism to work, it would be necessary that each hunting unit provides information on habitat measures implemented, in qualitative and quantitative terms (amount of surface contracted, length of hedges planted in m, etc.) to the competent authorities. It would be however important that the whole system is not too cumbersome or complicated in administrative terms for it to be efficient. This would allow the competent national authority to set priorities for allocation of hunting opportunities to each hunting ground and to distribute hunting opportunities accordingly, being mandatory to verify compliance with the hunting quota attributed to each unit or hunting ground. This information could be complemented by available information about the characteristics of the hunting area including habitat, information on breeding numbers or other variables that may be considered relevant. This system could be applied not only in the current AHM for the TD, but also in other game species to ensure conservation and wise use.

CRediT authorship contribution statement

Carlos Sánchez-García: Conceptualization, Data curation, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Thibaut Powolny:** Investigation, Writing – review & editing, Methodology. **Hervé Lormée:** Investigation, Methodology, Supervision, Validation, Writing – review & editing. **Susana Dias:** Investigation, Methodology, Writing – review & editing. **Francesc Sardà-Palomera:** Investigation, Methodology, Writing – review & editing. **Gerard Bota:** Investigation, Methodology, Writing – review & editing. **Beatriz Arroyo:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Carlos Sanchez Garcia-Abad reports financial support was provided by European Commission.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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